

**Dare to build the next generation of
realtime translation devices powered
by Edge AI**

Presented by Saigon AI Hub and Qualcomm

TECHNICAL PROPOSAL

Phase 2 — Technical Submission

Team / Project Name	[Your team name and project title]
Submission Date	[DD / MM / 2026]
Version	v1.0
Confidentiality	Restricted — Challenge Review Only

How to use this template

- **Fill out all sections:** Replace bracketed placeholders *[example]* with your response. Do not use the sample *italicized* text in your final submission.
- **Check the rubric:** Review the Evaluation Criteria on Page 2 before you start; judges evaluate strictly against this framework.
- **Highlight your advantage:** Dedicate extra effort to Section 3 (Business Solution) to clearly demonstrate why your approach is superior to existing alternatives.
- **Tailor your response:** Add any supplemental details or visual content that best showcases your solution.

Evaluation Criteria & Scoring Rubric

Read this before writing your proposal. Every section you complete maps to one or more criteria below.

Criterion	Weight	What Judges Look For
1. AI Approach & Technical Design	35%	Pipeline completeness, model choices, on-device optimisation strategy, accuracy targets and evaluation plan
2. Hardware & Device Concept	25%	Platform justification, BOM feasibility, power budget, form-factor suitability for the target environment
3. Business Solution	15%	How the solution addresses the industry gap, differentiation from existing tools, innovation strength, real-world applicability
4. Problem Definition & Impact	15%	Clarity of problem, real-world relevance, target user insight, design constraints identified
5. Team and Execution Plan	10%	Complementary skills, realistic timeline, milestone feasibility Early demo is a plus
TOTAL	100%	

 Your proposal is evaluated holistically. Strong AI design cannot compensate for missing business differentiation or weak hardware justification. Aim for depth and specificity in every section.

1. Executive Summary

Provide a 200–300 word overview of your entire proposal — the first section judges read.

 Cover: (1) the communication problem, (2) your solution and its business case, (3) target users, (4) key technical differentiator. 3–4 short paragraphs.

1.1 Problem Overview

[Describe the real-world voice communication gap — who is affected, where, and why existing tools fall short.]

1.2 Proposed Solution

[One or two sentences: solution name, core technology, deployment environment, and what makes it better than current alternatives.]

1.3 Key Value Proposition

[3–4 distinguishing capabilities versus existing solutions. Be specific and measurable.]

2. Problem Definition & Target Users

↗ Scoring weight: 15% — Problem Definition & Impact

2.1 Problem Statement

[Voice communication challenge: who is affected, the environment, why real-time translation is critical, and why current solutions fail to solve it.]

2.2 Impact Analysis

Impact Area	Current Pain Point	Consequence
Safety / Operations	[e.g., Misunderstood verbal safety instructions]	[e.g., Workplace incidents, compliance failures]
Productivity	[e.g., Translation delays between shifts]	[e.g., Downtime, rework, missed targets]
Connectivity	[e.g., No reliable internet in target environment]	[e.g., Existing translation apps unusable]

2.3 Target Users & Use Cases

User Segment	Language Need	Primary Context	Priority
[e.g., Local operators]	[e.g., VI ↔ EN]	[e.g., Assembly line, machine operations]	[e.g., Critical]
[e.g., Foreign supervisors]	[e.g., KO / ZH ↔ VI]	[e.g., Production management]	[e.g., Critical]
[e.g., Technical trainers]	[e.g., EN ↔ local lang.]	[e.g., Training, QA sessions]	[e.g., High]

2.4 Key Design Constraints

Constraint	Target / Requirement
Internet dependency	Zero — fully offline at runtime
End-to-end translation latency	[e.g., < 3 seconds]
Target deployment environment	[e.g., Industrial floor, clinic, checkpoint, field site]
Language pairs required	[e.g., VI ↔ EN, VI ↔ KO]
Additional constraints	[e.g., ruggedised housing, >8 h battery, no display required]

3. Business Solution & Innovation

↗ Scoring weight: 15% — Business Solutions

3.1 Industry Problem & Solution Fit

[Name 1–2 existing tools or approaches. For each, state the specific failure mode — latency, connectivity requirement, cost, or domain accuracy — and show how your solution directly addresses that gap.]

3.2 Innovation & Competitive Strengths

Dimension	Existing Solutions	Your Solution
Connectivity	[e.g., Cloud-required; fails in offline or low-bandwidth environments]	[e.g., Fully offline — no internet required at runtime]
Latency	[e.g., >5 s round-trip for cloud apps]	[e.g., <3 s end-to-end on-device]
Domain Accuracy	[e.g., Generic models; poor on technical or industry vocabulary]	[e.g., Fine-tuned on [industry] glossary; custom pronunciation lexicon]
Data Privacy	[e.g., Audio and text transmitted to external cloud servers]	[e.g., All processing on-device; no data leaves the hardware]

📌 Be specific — vague claims like "faster" or "more accurate" score low. Quantify the gap: e.g., "Cloud apps return in >5 s; our solution in <3 s at zero connectivity."

[In 2–3 sentences, summarise what makes your solution distinctly innovative compared with all available alternatives today.]

4. AI Approach & Technical Design

↗ Scoring weight: 35% — Highest weighted criterion

4.1 System Pipeline Overview

Describe your end-to-end AI pipeline and justify each stage. Include a pipeline diagram in your final submission.

[Map your pipeline stages — e.g., Audio Capture → VAD → ASR → NMT → TTS → Audio Output. Teams using end-to-end multilingual models or LLM-based approaches should describe their architecture equivalently.]

4.2 Module-by-Module Design

Module	Model / Framework	Size (est.)	Latency Target	Key Technique
<i>[e.g., VAD]</i>	<i>[e.g., Silero-VAD]</i>	<i>[~1 MB]</i>	<i>[< 10 ms/frame]</i>	<i>[Noise suppression, streaming]</i>
<i>[e.g., ASR]</i>	<i>[e.g., SenseVoice-Small]</i>	<i>[~250 MB]</i>	<i>[< 500 ms]</i>	<i>[INT8 QNN, CTC chunked decode]</i>
<i>[e.g., NMT / LLM]</i>	<i>[e.g., NLLB-200 Distilled]</i>	<i>[~300 MB]</i>	<i>[< 1,000 ms]</i>	<i>[4-bit quant, beam search]</i>
<i>[e.g., TTS]</i>	<i>[e.g., VITS / MMS-TTS]</i>	<i>[~80 MB]</i>	<i>[< 700 ms]</i>	<i>[ONNX on NPU/HTP]</i>

4.3 On-Device Optimisation & Memory Management

[Describe your quantisation approach and memory management strategy (streaming, lazy loading). State your target RAM ceiling and how you stay within it.]

4.4 Robustness & Edge Case Handling

[Describe how the system handles: (a) acoustic challenges — noise, echo, multiple speakers; (b) linguistic challenges — dialects, accents, code-switching; (c) edge cases — silence, rapid speech, out-of-vocabulary terms.]

5. Hardware & Device Concept

↗ Scoring weight: 25% — Hardware & Device Concept

5.1 Platform Selection & Justification

Compare at least one SoC / compute platform options and justify your choice.

Criteria	[Option A — name]	[Option B — name]	Selected?
NPU Performance (TOPS)	[e.g., 12 TOPS]	[e.g., 45 TOPS]	[✓ / —]
Power Consumption (TDP)	[e.g., 5–7 W]	[e.g., 23 W]	[✓ / —]
RAM / Storage Support	[e.g., 8 GB LPDDR5]	[e.g., 32 GB LPDDR5x]	[✓ / —]
AI SDK / Toolchain	[e.g., QNN, SNPE]	[e.g., QNN, ONNX]	[✓ / —]
Form Factor Suitability	[e.g., Wearable/handheld]	[e.g., Portable device]	[✓ / —]

[In 2–3 sentences, justify your final platform choice and describe the device form factor — type (wearable, handheld, lanyard), dimensions, weight, and environmental rating.]

5.2 Key Hardware Components & Power Budget

Component	Specification	Peak Power	Notes
SoC Module	[e.g., QCS6490 Dev Kit]	[e.g., 5.0 W]	[CPU + NPU active]
Microphone Array	[e.g., 3× MEMS, 16 kHz]	[e.g., 0.1 W]	[Beamforming, noise-cancelling]
Speaker / Audio Out	[e.g., 3.5 mm + BT 5.3]	[e.g., 0.5 W]	
Battery	[e.g., 4,000 mAh Li-Po]	—	[Target: >8 h continuous]
Storage	[e.g., 64 GB UFS 3.1]	[e.g., 0.2 W]	[OS + all AI models]
Display (optional)	[e.g., 2.4" OLED / None]	[e.g., 0.3 W]	
Total System Budget		[e.g., ~6.1 W]	[Must sustain >8 h on target battery]

6. System Architecture & Integration

↗ This section bridges your AI pipeline (Section 4) and your hardware platform (Section 5). Judges assess whether the software and hardware layers work coherently as a complete system.

6.1 Software Stack

Layer	Component / Framework	Role
OS	[e.g., Android / Linux]	Base runtime & audio I/O
AI Runtime	[e.g., Qualcomm QNN SDK]	NPU inference execution
Audio Processing	[e.g., WebRTC / RNNoise]	Noise reduction & VAD
Model Serving	[e.g., ONNX Runtime / QNN]	ASR, NMT, TTS inference
App Layer	[e.g., Kotlin / C++ / Python]	UI, pipeline orchestration

6.2 Architecture Diagram Placeholder

[Insert Architecture Diagram Here]

6.3 Offline-First Design Principles (if any)

[e.g.,

1. All AI models stored on-device; no API calls at runtime
2. Language pack management: pre-load required language pairs on setup
3. Local caching of common phrase translations for sub-100 ms repeat responses
4. Fail-safe audio output: text-to-speaker fallback if TTS model load fails]

7. Team Profile & Project Timeline

↗ Scoring weight: 10% — Team and Execution Plan

7.1 Team Members

Name	Role	Expertise	Contribution Area
[Full Name]	[e.g., Team Lead / AI Engineer]	[e.g., NLP, On-device ML]	[e.g., ASR + NMT pipeline]
[Full Name]	[e.g., Hardware Engineer]	[e.g., Embedded Systems, SoC]	[e.g., BOM + power design]
[Full Name]	[e.g., ML Engineer]	[e.g., Model optimisation, QNN]	[e.g., Quantisation + NPU tuning]
[Full Name]	[e.g., UX / Product Designer]	[e.g., UI/UX, User Research]	[e.g., Form factor + UAT]

7.2 Project Timeline

Phase	Milestone	Key Activities	Target Date
1	Requirements & Research	[Problem validation, platform selection, model survey]	[DD/MM/YYYY]
2	Model Selection & Optimisation	[Model selection, quantisation, initial benchmarks]	[DD/MM/YYYY]
3	On-Device Integration & Testing	[SDK deployment, pipeline integration, WER/BLEU/MOS/UAT]	[DD/MM/YYYY]
4	Finalisation & Submission	[Write-up, demo video, packaging, export PDF + DOCX]	[DD/MM/YYYY]

8. Submission Checklist

All items must be completed before final submission. Missing sections score zero on the relevant rubric criterion.

#	Checklist Item	Status
1	Executive Summary written (200–300 words)	☐ Done
2	Problem Statement, Target Users, and Design Constraints completed	☐ Done
3	Business Solution completed — industry gap, solution fit, and competitive differentiation described	☐ Done
4	AI pipeline documented with model specs, latency targets, and optimisation strategy	☐ Done
5	Hardware platform justified; form factor, BOM and power budget filled in	☐ Done
6	Team profiles and project timeline filled in	☐ Done
7	All placeholder text replaced; pipeline/architecture diagram inserted (Section 6.2)	☐ Done
8	Demo video or prototype link attached (if available); document exported as .PDF	☐ Done